MALL-PE-08: Major Hardware/Environmental Fault Resolution - Cisco PE Role

Node: MALL-PE-08 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_6530

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution procedures for a major hardware/environmental fault occurring on the Cisco PE role at the Malleswaram Node (Node ID: MALL-PE-08). The failure domain is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, as indicated by the ALARM_6530 alarm code of severity major. The underlying protocol and hardware mechanics responsible for this failure domain can be attributed to the complex interplay of ASIC memory parity checking, line card redundancy protocols, and power supply redundancy mechanisms inherent to the Cisco PE equipment. Specifically, the PE role relies on the Advanced Services Interface Card (ASIC) to manage high-speed traffic, and the line card redundancy protocol ensures continuous service availability in case of a hardware failure. However, when the ASIC memory parity checking mechanism fails, it can lead to line card crashes and power supply redundancy loss, resulting in the ALARM_6530 alarm code. This SOP is designed to guide the resolution of such a failure domain by identifying the root causes, executing diagnostic commands, implementing resolution steps, verifying the fix, and rolling back if necessary.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Malleswaram Node to perform physical checks and troubleshooting.

- Verify the redundant link status on the PE equipment to ensure continuous service availability.

- Perform a site survey to identify any environmental factors that may be contributing to the hardware fault.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: show asp drop, Expected Output: ASIC drop statistics, Description: This command isolates the ASIC drop statistics, which can indicate ASIC memory parity errors or other issues that may be contributing to the hardware fault.
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```
Command: show redundancy redundancy, Expected Output: Redundancy status, Description: This command verifies the redundancy status of the line card and power supply, which can indicate potential issues with redundancy protocols.

Command: show environment temperature, Expected Output: Temperature reading, Description: This command isolates the temperature reading of the PE equipment, which can indicate environmental factors that may be contributing to the hardware fault.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: hw-module subslot 0/0 replace, Description: This command replaces the faulty ASIC with a new one, which can resolve ASIC memory parity errors and other issues related to the ASIC.

Step 2: Step: Reboot the PE equipment, Command: reload, Description: This command reboots the PE equipment, which can resolve line card crashes and other issues related to the line card redundancy protocol.

Step 3: Step: Verify power supply redundancy status, Command: show redundancy redundancy, Description: This command verifies the redundancy status of the power supply, which can ensure that the PE equipment is operating within the specified redundancy protocol.

Post-Check Verification

```
Command: show asp drop, Expected Output: ASIC drop statistics (resolved), Description: This command verifies that the ASIC drop statistics are resolved, indicating that the ASIC memory parity error is fixed.

Command: show redundancy redundancy, Expected Output: Redundancy status (resolved), Description: This command verifies that the redundancy status of the line card and power supply is resolved, indicating that the line card crash and power supply redundancy loss are fixed.

Command: show environment temperature, Expected Output: Temperature reading (within normal range), Description: This command verifies that the temperature reading of the PE equipment is within the normal range, indicating that the environmental factors contributing to the hardware fault are resolved.
```

Roll-back Procedures

- Step: Restore the original ASIC configuration, Command: copy tftp:/// startup-config, Description: This command restores the original ASIC configuration from the TFTP server, which can roll back the changes made during the resolution process.

- Step: Reboot the PE equipment, Command: reload, Description: This command reboots the PE equipment, which can roll back the changes made during the resolution process and restore the original state of the equipment.

- Step: Verify the original state of the equipment, Command: show running-config, Description: This command verifies that the equipment is restored to its original state, indicating that the rollback procedure was successful.